

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

Using Large-Scale ACN Charging Session Data

14,299

Sessions Analysed

3 Agents

Agentic AI Pipeline

752 hrs

Evaluation Window

96.3%

User Satisfaction

01 | Data Landscape & Preprocessing

ACN-Data (Caltech/JPL)

14,299 sessions · JSON → CSV · Timestamps, kWh, station IDs, user behavior

UrbanEV (ST-EVCDP)

24,798 charging piles · 5-min intervals · Informed 80% surge & 30% discount threshold selection (not directly integrated — geographic & format incompatibility with ACN-Data documented)

Engineered Features

- Charger Utilization Rate (Charging Time / Total Available Time)
- Revenue per Session & Energy Cost per kWh
- Queue Length Proxy & Occupancy Density
- Hourly aggregation with day-of-week & hour features

Preprocessing Pipeline

01

JSON Repair & Conversion

Malformed ACN JSON fixed in-memory → converted to structured CSV

02

Datetime Standardisation

All timestamps converted to UTC-aware datetime objects

03

Missing Value Handling

Transparent imputation with documented assumptions at each stage

04

Hourly Aggregation

Sessions aggregated to session_count, kwh_demand, utilization per hour

05

Train / Eval Split

4,463 hrs training · 752 hrs evaluation (last 31 days)

02 | Key EDA Findings & Demand Behaviour

Peak Demand Windows

Charging demand peaks between 08:00–10:00 and 17:00–19:00, aligning with workplace commute patterns at Caltech/JPL sites

Weekday vs Weekend

Weekday sessions are 3× higher than weekends — fleet and workplace charging dominate. Weekend demand is flatter and more price-elastic

Energy Distribution

Median session delivers ~7.47 kWh with high right-skew. 20% of sessions deliver 60%+ of total energy — key targets for dynamic pricing

Utilization Volatility

Off-peak hours (00:00–06:00) show <10% utilization. Average evaluation utilization: 37.5% — significant headroom for off-peak incentives

03 | Demand Prediction Agent — Modelling & Results

Model Architecture

- Historical lookup table conditioned on day-of-week × hour-of-day
- Trained on 4,463 hourly samples (April – November 2018)
- Outputs: predicted_kwh, predicted_sessions per time slot
- Evaluated on last 752 hours (≈31 days) of data
- Features: session_count, kwh_demand, dayofweek, hour

Model Performance

17.59 kWh MAE — Mean Absolute Error

27.47 kWh RMSE — Root Mean Sq. Error

0.293 R^2 R^2 Score — Variance Explained

Interpretation

R^2 of 0.293 reflects an intentional rule-based baseline — the lookup table captures core temporal patterns but underestimates high-variance peaks. MAE of 17.59 kWh measures hourly aggregate demand error across all stations, not per-session error. The agentic architecture is model-agnostic; substituting gradient boosting or LSTM is the primary future improvement direction.

Tariff Pricing Logic

Utilization > 80%

Surge pricing — tariff increases above base rate

Utilization < 30%

Discount signal — tariff drops to incentivise off-peak use

30% – 80% range

Dynamic adjustment based on predicted vs. average kWh demand

Off-Peak Analysis

411 off-peak hours identified (utilization < 30%) — avg tariff \$0.246/kWh applied as discount signal to uplift sessions during low-demand windows

Revenue Outcomes

+\$230.47

Net Revenue Lift

+6.89%

05 | Monitoring & Learning Agent — Feedback Loop

13,388.59kWh

Total Energy Delivered

37.5%

Demand Fulfillment Rate

96.3%

User Satisfaction Score

Feedback Loop Performance

Observe

Agent monitors actual kWh delivered, session counts, and tariff applied per hour across all 752 evaluation steps

Evaluate

Computes revenue lift vs flat baseline, utilization achieved vs target, and user satisfaction derived from tariff deviation

Adapt

Calibrates pricing sensitivity parameters — widens discount window when utilization falls below 30%, tightens surge threshold at 80%+

Report

Generates summary metrics: MAE 17.59, RMSE 27.47, Net Revenue Lift +\$230.47, Avg Satisfaction 96.3%

06 | Business, Operational & Policy Implications

Business

- +6.89% revenue lift over flat-rate pricing with zero infrastructure change
- Dynamic pricing captures demand elasticity — higher willingness to pay at peak
- Pricing Efficiency Score: \$0.267/kWh delivered vs \$0.25 flat baseline

Operational

- 37.5% demand fulfillment rate — off-peak discounts can shift 411 hrs of demand
- 96.3% user satisfaction maintained despite dynamic pricing adjustments
- Agentic feedback loop continuously refines tariff decisions without manual intervention

Policy

- Surge pricing above 80% utilization can reduce peak-hour congestion and queue times
- Off-peak incentives align with grid demand-response objectives
- Transparent assumption logging supports regulatory compliance and auditability

Limitations & Future Work

$R^2=0.293$ reflects intentional rule-based baseline — model-agnostic architecture supports future ML substitution. MAE=17.59 kWh is hourly aggregate error across all stations, not per-session. Satisfaction and utilization are proxy metrics — no direct user feedback or time-based charger data available. UrbanEV informed thresholds only; direct integration is future work. No causal claims — all findings are correlational.